



Chapter 2

Limits and Continuity

MAE101 - CALCULUS

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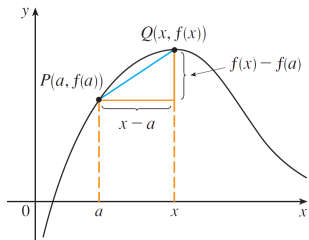
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The tangent problem

1 The limit of a function

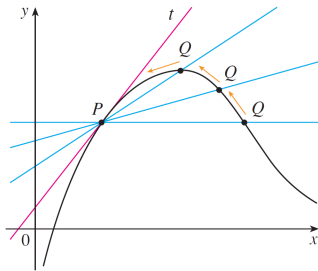


- A curve C has equation $y = f(x)$ and we want to find the tangent line to C at the point $P(a, f(a))$,
- We consider a nearby point $Q(x, f(x))$, where $x \neq a$.
- The slope of the secant line PQ :

$$m_{PQ} = \frac{f(x) - f(a)}{x - a}$$

The tangent problem

1 The limit of a function



- We let Q approach P along the curve C by letting x approach a ,
- If m_{PQ} approaches a number m , then we define the tangent t to be the line through P with slope m .

$$m = \lim_{Q \rightarrow P} m_{PQ} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

The limit of a function

1 The limit of a function

Definition: We write

$$\lim_{x \rightarrow a} f(x) = L$$

and say the limit of $f(x)$, as x approaches a , equals L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a but not equal to a .

Proposition: The limit of a function is unique if it exists.

Example.

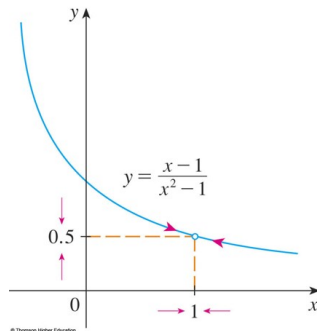




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One-Sided Limits

2 One-Sided Limits

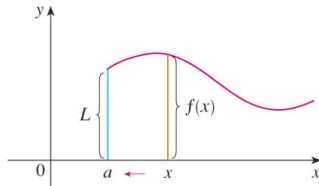
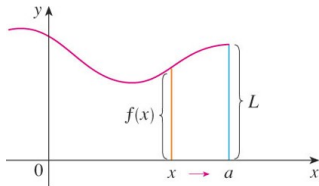
We write

$$\lim_{x \rightarrow a^-} f(x) = L$$

and say the left-hand limit of $f(x)$ as x approaches a equals L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a **but not equal to a and x less than a .**

Similarly, the right-hand limit of $f(x)$ as x approaches a equals L and we write

$$\lim_{x \rightarrow a^+} f(x) = L$$





Relating One-Sided and Two-Sided Limits

2 One-Sided Limits

Theorem

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L$$

Example. Prove that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

Relating One-Sided and Two-Sided Limits

2 One-Sided Limits

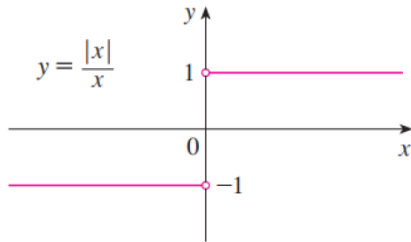
Theorem

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L$$

Example. Prove that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

Solution.

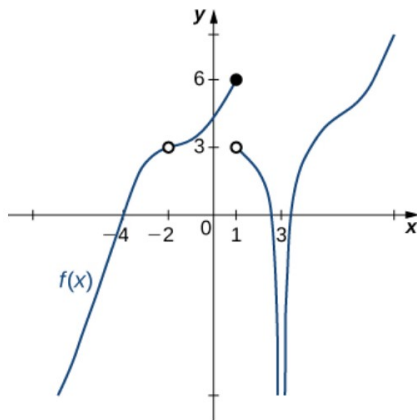
$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{|x|}{x} &= \lim_{x \rightarrow 0^+} \frac{x}{x} = 1 \\ \lim_{x \rightarrow 0^-} \frac{|x|}{x} &= \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1. \end{aligned}$$



Example

2 One-Sided Limits

Use the graph of $f(x)$ in to determine each of the following values:

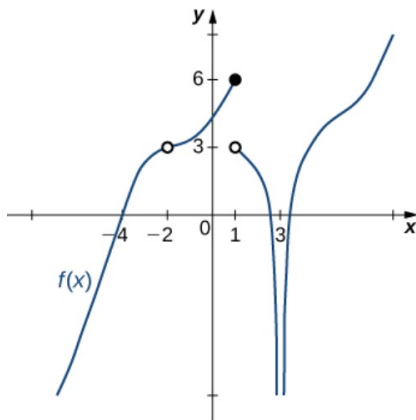


- $\lim_{x \rightarrow -4^-} f(x), \lim_{x \rightarrow -4^+} f(x), \lim_{x \rightarrow -4} f(x).$
- $\lim_{x \rightarrow -2^-} f(x), \lim_{x \rightarrow -2^+} f(x), \lim_{x \rightarrow -2} f(x).$
- $\lim_{x \rightarrow 1^-} f(x), \lim_{x \rightarrow 1^+} f(x), \lim_{x \rightarrow 1} f(x).$
- $\lim_{x \rightarrow 3^-} f(x), \lim_{x \rightarrow 3^+} f(x), \lim_{x \rightarrow 3} f(x).$

Example

2 One-Sided Limits

Use the graph of $f(x)$ in to determine each of the following values:



a. $\lim_{x \rightarrow -4^-} f(x)$, $\lim_{x \rightarrow -4^+} f(x)$, $\lim_{x \rightarrow -4} f(x)$.

b. $\lim_{x \rightarrow -2^-} f(x)$, $\lim_{x \rightarrow -2^+} f(x)$, $\lim_{x \rightarrow -2} f(x)$.

c. $\lim_{x \rightarrow 1^-} f(x)$, $\lim_{x \rightarrow 1^+} f(x)$, $\lim_{x \rightarrow 1} f(x)$.

d. $\lim_{x \rightarrow 3^-} f(x)$, $\lim_{x \rightarrow 3^+} f(x)$, $\lim_{x \rightarrow 3} f(x)$.

Solution.

a. $\lim_{x \rightarrow -4^-} f(x) = 0$, $\lim_{x \rightarrow -4^+} f(x) = 0$, $\lim_{x \rightarrow -4} f(x) = 0$.

b. $\lim_{x \rightarrow -2^-} f(x) = 3$, $\lim_{x \rightarrow -2^+} f(x) = 3$, $\lim_{x \rightarrow -2} f(x) = 3$.

c. $\lim_{x \rightarrow 1^-} f(x) = 6$, $\lim_{x \rightarrow 1^+} f(x) = 3$, $\lim_{x \rightarrow 1} f(x)$ does not exist.

d. $\lim_{x \rightarrow 3^-} f(x) = -\infty$, $\lim_{x \rightarrow 3^+} f(x) = -\infty$,



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- ▶ Problems

Limit laws

3 Limit laws

Theorem

Suppose that c is a constant and the limit $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exists. Then

1. $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
2. $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$
3. $\lim_{x \rightarrow a} cf(x) = c \lim_{x \rightarrow a} f(x)$
4. $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$
5. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$ if $\lim_{x \rightarrow a} g(x) \neq 0$
6. $\lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n, \forall n \in \mathbb{N}$
7. $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$
 $(\forall \lim_{x \rightarrow a} f(x) \text{ if } n \text{ is odd and for } \lim_{x \rightarrow a} f(x) \geq 0 \text{ if } n \text{ is even and } f(x) \geq 0).$

Example. a. $\lim_{x \rightarrow -3} 4x + 2$

b. $\lim_{x \rightarrow 2} \frac{2x^2 - 3x + 1}{x^3 + 4}$

c. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$



The Squeeze Theorem

3 Limit laws

The Squeeze Theorem

Let $f(x)$, $g(x)$, and $h(x)$ be defined for all $x \neq a$ over an open interval containing a . If

$$f(x) \leq g(x) \leq h(x)$$

for all $x \neq a$ in an open interval containing a and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

where L is a real number, then $\lim_{x \rightarrow a} g(x) = L$.

Example. Evaluate the following limits $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$



L'Hospital's Rule

3 Limit laws

L'Hospital's Rule

Suppose that we have one of following cases, $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$ where a can be any real number, infinity or negative infinity. In these case we have,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Example. Evaluate the following limits

a. $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

b. $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$



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Horizontal Asymptotes

4 Asymptotes

Definition

If $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$, we say the line $y = L$ is a **horizontal asymptote** of f .

Example. Given $f(x) = 5 - \frac{2}{x^2}$, evaluate $\lim_{x \rightarrow \infty} f(x)$, $\lim_{x \rightarrow -\infty} f(x)$ and determine the horizontal asymptote(s) for f .

Horizontal Asymptotes

4 Asymptotes

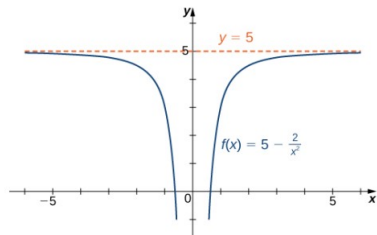
Definition

If $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$, we say the line $y = L$ is a **horizontal asymptote** of f .

Example. Given $f(x) = 5 - \frac{2}{x^2}$, evaluate $\lim_{x \rightarrow \infty} f(x)$, $\lim_{x \rightarrow -\infty} f(x)$ and determine the horizontal asymptote(s) for f .

Solution. Using the algebraic limit laws, we have $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = 5$.

Therefore, $f(x) = 5 - \frac{2}{x^2}$ has a horizontal asymptote of $y = 5$ and f approaches this horizontal asymptote as $x \rightarrow \pm\infty$ as shown in the following graph.





Vertical asymptotes

4 Asymptotes

Definition

Let $f(x)$ be a function. If any of the following conditions hold, the line $x = a$ is a **vertical asymptotes** of $f(x)$:

$$\lim_{x \rightarrow a} f(x) = +\infty, \quad \lim_{x \rightarrow a^-} f(x) = +\infty, \quad \lim_{x \rightarrow a^+} f(x) = +\infty, \quad \lim_{x \rightarrow a} f(x) = -\infty, \quad \lim_{x \rightarrow a^-} f(x) = -\infty, \quad \lim_{x \rightarrow a^+} f(x) = -\infty$$

Example. Evaluate each of the following limits using Infinite Limits from Positive Integers. Identify any vertical asymptotes of the function $f(x) = 1/x$

a. $\lim_{x \rightarrow 0} f(x)$ b. $\lim_{x \rightarrow 0^-} f(x)$ c. $\lim_{x \rightarrow 0^+} f(x)$

Vertical asymptotes

4 Asymptotes

Definition

Let $f(x)$ be a function. If any of the following conditions hold, the line $x = a$ is a **vertical asymptotes** of $f(x)$:

$$\lim_{x \rightarrow a} f(x) = +\infty, \quad \lim_{x \rightarrow a^-} f(x) = +\infty, \quad \lim_{x \rightarrow a^+} f(x) = +\infty, \quad \lim_{x \rightarrow a} f(x) = -\infty, \quad \lim_{x \rightarrow a^-} f(x) = -\infty, \quad \lim_{x \rightarrow a^+} f(x) = -\infty$$

Example. Evaluate each of the following limits using Infinite Limits from Positive Integers. Identify any vertical asymptotes of the function $f(x) = 1/x$

a. $\lim_{x \rightarrow 0} f(x)$ b. $\lim_{x \rightarrow 0^-} f(x)$ c. $\lim_{x \rightarrow 0^+} f(x)$

Solution. $\lim_{x \rightarrow 0^-} f(x) = -\infty,$

$$\lim_{x \rightarrow 0^+} f(x) = +\infty, \quad \lim_{x \rightarrow 0} f(x) \text{ DNE.}$$

The function $f(x) = 1/x$ has a vertical asymptote of $x = 0$.

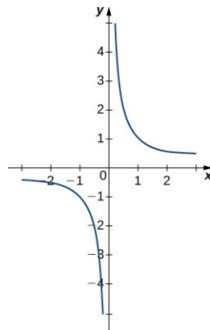




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Continuity

5 Continuity of a function

Definition

A function $f(x)$ is **continuous at a point a** if and only if the following three conditions are satisfied:

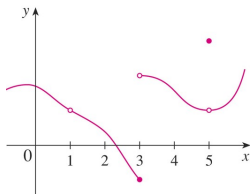
i. $f(a)$ is defined

ii. $\lim_{x \rightarrow a} f(x)$ exists

iii. $\lim_{x \rightarrow a} f(x) = f(a)$

A function is **discontinuous at a point a** if it fails to be continuous at a .

Example 1. At which numbers is f discontinuous? Why?



Example 2. Determine whether the function

$$f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

is continuous at $x = 0$.



Continuity

5 Continuity of a function

Continuity from the Right and from the Left

A function $f(x)$ is said to be **continuous from the right** at a if $\lim_{x \rightarrow a^+} f(x) = f(a)$.

A function $f(x)$ is said to be **continuous from the left** at a if $\lim_{x \rightarrow a^-} f(x) = f(a)$.

Definition

A function f is continuous on an interval if it is continuous at every number in the interval.

Composite Function Theorem

If $f(x)$ is continuous at L and $\lim_{x \rightarrow a} g(x) = L$, then $\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x)) = f(L)$

Theorem

If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g(x) = f(g(x))$ is continuous at a .



Theorem

5 Continuity of a function

Theorem

If f and g are continuous at a and c is a constant, then the following functions are also continuous at a :

1. $f + g$
2. $f - g$
3. cf
4. fg
5. $\frac{f}{g}$ if $g(a) \neq 0$.

Theorem

The following types of functions are continuous at every number in their domains:

1. Polynomials
2. Rational functions
3. Root functions
4. Trigonometric functions

Example. State the interval(s) over which the function $f(x) = \frac{x - 1}{x^2 + 2x}$ is continuous.



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6 The Intermediate Value Theorem

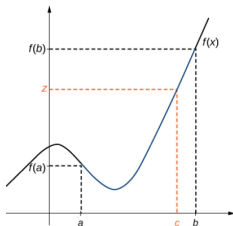
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The Intermediate Value Theorem

6 The Intermediate Value Theorem

Theorem

Let f be continuous over a closed, bounded interval $[a, b]$. If z is any real number between $f(a)$ and $f(b)$, then there is a number c in $[a, b]$ satisfying $f(c) = z$.



Example. Check whether there is a solution to the equation $f(x) = x^5 - 2x^3 - 2$ between the interval $[0; 2]$.

Figure: There is a number $c \in [a, b]$ that satisfies $f(c) = z$.

The Intermediate Value Theorem

6 The Intermediate Value Theorem

Theorem

Let f be continuous over a closed, bounded interval $[a, b]$. If z is any real number between $f(a)$ and $f(b)$, then there is a number c in $[a, b]$ satisfying $f(c) = z$.

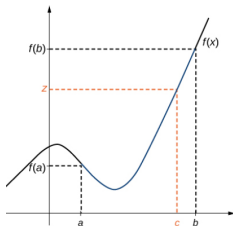


Figure: There is a number $c \in [a, b]$ that satisfies $f(c) = z$.

Example. Check whether there is a solution to the equation $f(x) = x^5 - 2x^3 - 2$ between the interval $[0; 2]$.

Solution. We have $f(0) = -2 < 0$ and $f(2) = 14 > 0$.

Thus, $f(0) < 0 < f(2)$, that is $z = 0$ is a number between $f(0)$ and $f(2)$.

So, the IVT says there is a number $c \in [0, 2]$ such that $f(c) = 0$. In other words, the equation $x^5 - 2x^3 - 2 = 0$ has at least one root $c \in [0, 2]$.



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Problems

7 Problems

1. Evaluate the following limits

a. $\lim_{x \rightarrow 1} \frac{x^6 - 1}{x^{10} - 1}$

b. $\lim_{x \rightarrow 0} \frac{\tan 3x + 2x}{\tan 5x - \sin x}$

c. $\lim_{x \rightarrow \infty} \frac{x^2 + x - 12}{x^3 - 3}$

d. $\lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right)$

e. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{|x - 1|}$

2. Find the constant m that makes f continuous on its domain

a. $f(x) = \begin{cases} x^2 - m^2, & x < 4 \\ mx + 20, & x \leq 4 \end{cases}$

b. $f(x) = \begin{cases} mx^2 + 2x, & x < 2 \\ x^3 - mx, & x \geq 2 \end{cases}$

3. Find the numbers at which the function $f(x) = \begin{cases} x + 2, & x < 0 \\ 2x^2, & 1 \geq x \geq 0 \\ 2 - x, & x > 1 \end{cases}$ is

discontinuous.



Q&A

Thank you for listening!